

AUTOFLOW

Support That Runs Itself.

Resolves Routine Order Tickets. Escalates The Rest.

DallasAI Summer Program

Group 7

The Same Tickets. All Day.

Routine Order Requests Can Consume Significant Support-Team Time.

Order Status

"Where's My Order."

Cancellations

"Cancel My Order."

Returns

"I Want To Return This."

Up To

60%

Of Support Tickets Are The Same Order Questions

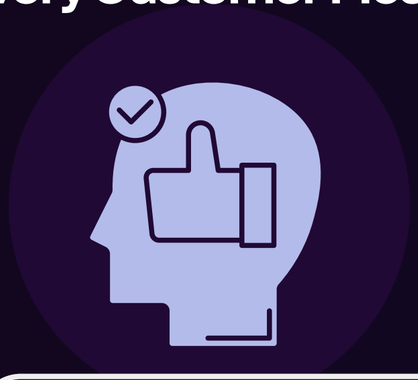
Manual Workflow

- Find Order
- Check Policy
- Take Action
- Reply

Chatbots Answer. Support Teams Still Complete The Work.

Resolves The Routine. Escalates The Rest.

Every Customer Message Flows Through Four Steps.



Understand

Classify The Intent
Pull The Order ID and SKU.



Act

Call Typed Tools Connected To The Live Store. Never Guessing From Training Data.



Prove

Ground Every Reply In Written Policy And Show The Tools It Used.



Hand Off

Out Of Scope Or Blocked By A Rule? Escalate To A Human, Don’t Bluff.

Order status · Returns · Wrong / Defective Items · Cancellations

Rules in The Backend. Reasoning on Surface.

Simplified Architecture of AutoFlow.

Customer Chat



Streaming Replies + Collapsible Reasoning Trace

AutoFlow Agent



Intent Understanding + Orchestrated Reasoning

Store Systems



Tools & Integrations + Policy Retrieval

Resolve / Escalate



Approve / Edit / Send; Escalate When Needed



Business Rules Live In The Backend, Not the Prompt

Bounded. Grounded. Traceable.

The Architecture Behind Reliable Automation.

Customer

React + Vite Chat UI.
Streams Every Turn.

Single Source of Truth

Business rules live in the backend, never in the prompt.

AGENT * LangGraph State Machine

Intake
Classifying Intent + Extract Slots

Router
Applies 5 Escalation Rules

Handlers
Call Typed Tools Per Intent

Backend * FastAPI +SQLite

Typed REST Tools Over HTTP.
The Mock Store The Agent Acts On.

LLM

Llama 3.3
Groq + Fallback

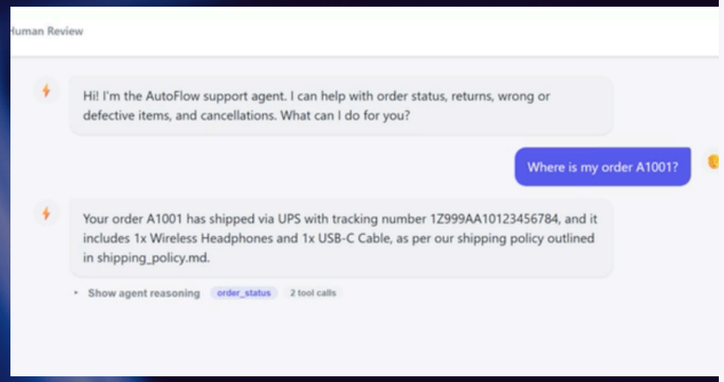
Retrieval

Chroma
Policy Docs

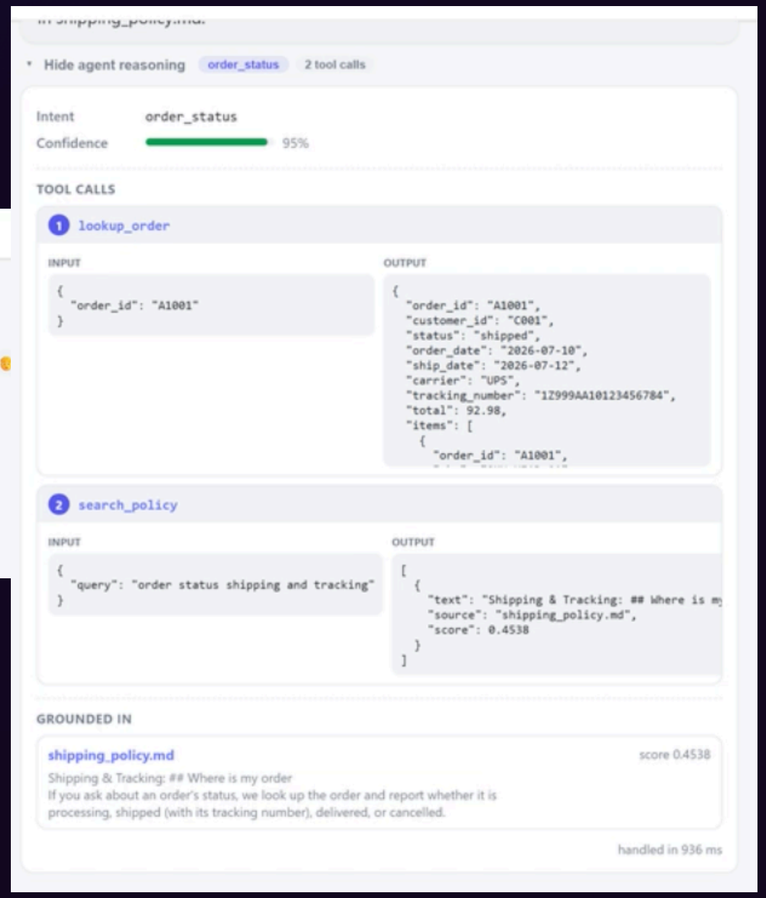
Every Step Is Logged

See It Reason.

Resolved Request.



Reasoning Trace.



Intent · Confidence · Tools · Source

1 Second
Median response per turn

37 Cases
Evaluated

100%
Task Completion

It Knows When To Stop.

Escalate When Needed. Protect Customers. Keep Humans In Control.



Cancel Order

AI Attempts Action



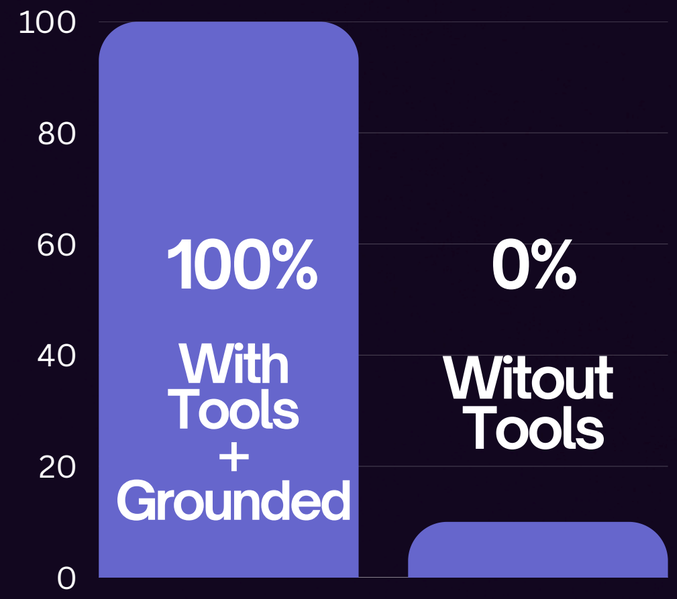
Order Already Shipped

Action Blocked



Human Review

Reply Returns



Protects Customers **Preserves Trust** **Keeps Humans in Control** **Closes The Loop**

30 Hours Saved/Week

Estimated Weekly Time Savings.

500	×	60%	×	6	=	30
Tickets/Week		Supported		Minutes		Hours Saved

6 Months Pilot Partners

12 Months More flows & Analytics

24 Months Governed Verticals

Our Ask

Pilot Partners
Feedback

Practical Agentic Workflow Automation, Without The Enterprise Price Tag.

Our Team



Farzana Lakhani



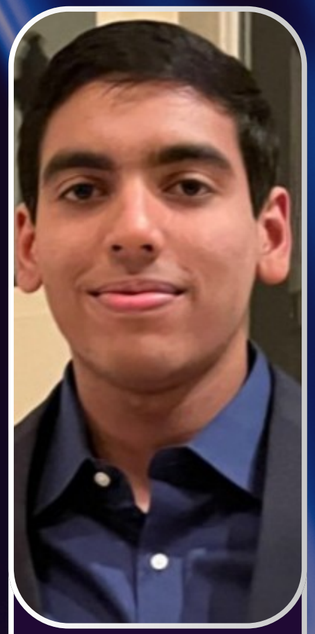
Malcolm T. Gomez



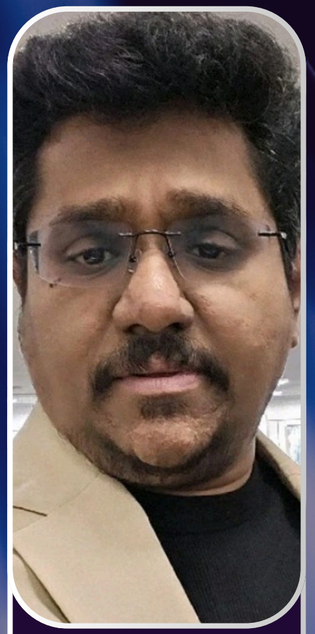
Sarow Saeedi



Leticia Molina

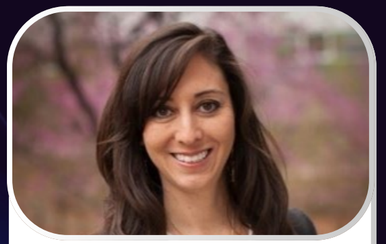


Sparsh Mathur



Thiyagarajan Kamalakannan

Mentors



Beckie Callahan



Ashrit Kulkarni

Thank You!